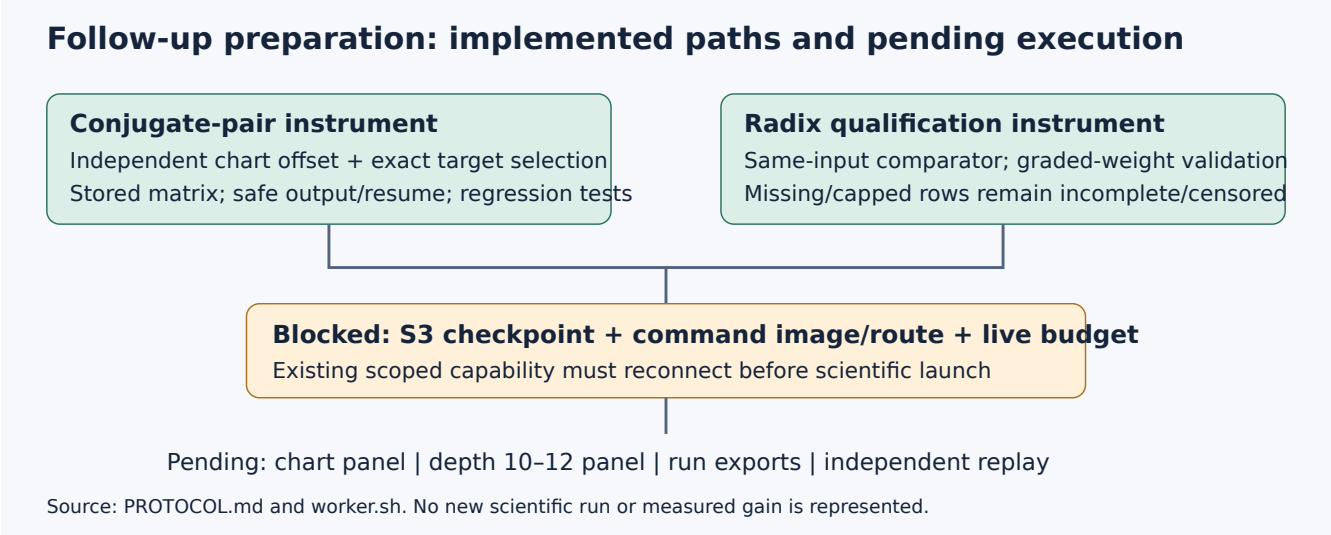
Conjugate-pair repair and radix-grading follow-up

November 6, 2026 · Development report

Native chart selection/resume support and a matched-input radix comparator are implemented. Scientific qualification panels remain blocked and have not launched. Development observations are producer assertions under PCC-20261106-7ccd3b, standing unverified.



Conjugate-pair continuation

The instrument accepts independent chart offsets and exact target filters, stores the actual matrix and solver seed, preserves the historical seed-zero stream, and resumes interrupted pairs without overwriting earlier output. Tests cover same-file append, separate-file history, malformed/truncated input, duplicate identities and seed/degree separation. The prospective panel uses two original targets, three chart offsets and plain/transformed controls. Correctness and solver completion are separate outputs; an empty set is not a successful solve merely because it matches an empty oracle.

Radix continuation

The native comparator checks equal curves, tower parameters, targets, construction seeds, budgets and grading weights. Missing draws are refused; capped runs retain unknown ratios. The prospective panel compares radix 2/4/8 at depths 10/11/12 with tower/null controls, two draws and four threads. Weighted degree values are not compared as costs. The existing depth-10 data provides a compatibility check, not a new trial.

Execution dependency

The mandatory S3 checkpoint gate has no upload transport in this runtime. An authenticated command-capable EKS route, exact image/source/binary binding and live budget/capacity admission remain required. The ordinary GitHub experiment dispatch accepts ecbench specifications and is not a substitute for this example payload. No scientific computation is running in the background.

Evidence and next step

The complete populations, controls, deadlines, accounting exclusions and stop conditions are in PROTOCOL.md. The blocked backlog cards are XP-20261106-conjugate-chart-followup and XP-20261106-radix-depth-followup. The worker script is a payload, not a submitter. Development commands and actual outcomes are in validation/checks.md and validation/native-tests.txt. A preparation PR does not complete either experiment.

Sources checked and limits

KB unavailable: no tools or executable/catalog; the repository wrapper exits 127. Direct sources: the two existing conjugate-pair studies; pkm_radix_grading_20261104; gaudry_quartic.rs; Petit–Kosters–Messeng, PKC 2016, sections 2.4, 3, 4.2 and 5, https://people.maths.ox.ac.uk/petit/files/16PKC_primeECDLP.pdf. The primary Gaudry PDF at eprint 2004/027 returned HTTP 403. No new theorem, novelty claim or measured gain is asserted. Historical scientific plots and dashboards remain unchanged because no new scientific value exists.